Nonparametric Predictive Inference for the Single-Period Inventory Model

Kholood O. Alyazidi[a,b], Frank P.A. Coolen[a] and Tahani Coolen-Maturi[a]

[a]Department of Mathematical Sciences, Durham University, Durham, UK; [b]Statistics and Operations Research Department, King Saud University, Riyadh, Saudi Arabia

ARTICLE HISTORY
Compiled September 23, 2025

ABSTRACT
In inventory theory, a known probability distribution is traditionally assumed for the random demand. In this paper, an alternative approach to inventory problems is presented, with the aim of basing the order strategy on information in the form of previously observed demands, adding only quite minimal further assumptions. Nonparametric Predictive Inference (NPI) is used to predict a future demand given observations of past demands. NPI makes only few modelling assumptions, which is achieved by quantifying uncertainty through lower and upper probabilities. As the first use of NPI in inventory theory, the basic scenario of inventory for a single period is considered. The performance of the NPI approach is investigated through simulations, which are also used to compare the method to the classical approach in which the probability distribution of the random demand is assumed to be known. Several cases are studied, some where the assumptions underlying the classical method are fully correct and other cases where the assumed model is not well aligned with the reality. The NPI approach performs well, already outperforming the classical method for relatively small data sets if there is substantial discrepancy between the classical method assumptions and reality.

KEYWORDS
Demand; inventory; lower and upper expected profits; lower and upper probabilities; Nonparametric Predictive Inference (NPI); single period.

1. Introduction

Inventory theory is one of the main areas in Operational Research, with a long history of contributions to the literature in which many scenarios are studied. The expression 'inventory' is typically used to define the total number of goods or materials contained in a store or factory at any given time (Karteek & Jyoti, 2014). Common problems for an inventory system include the stocking of spare parts, perishable items and seasonal items. Inventories need to be carefully managed to avoid problems and to determine and control the associated costs and risks of shortages, which affect a company's profit or loss (Reid & Sanders, 2007).

Harris (1913) presented the first inventory model in the literature. Many researchers have extended Harris's model, considering different types of demand and replenishment. According to Hadley and Whitin (1963), the inventory models are classified

CONTACT Frank Coolen, Email: frank.coolen@durham.ac.uk

based on the pattern of demand, which may be either deterministic or random over time. Deterministic demand occurs if the quantities needed over a period of time are exactly known. Random demand occurs when the demand is not known within a period of time. Forecasting the random demand is considered to be a major difficulty faced by the decision maker.

In the field of Operational Research, the examination of inventory problems traditionally involves assuming a known probability distribution for the random demand. This classical approach varies the number of periods, or decision moments, and incorporates associated costs. Recently, different approaches for dealing with uncertainties have been applied to inventory problems, including robust optimisation and fuzzy logic. For example, Chu, Huang, and Thiele (2019) introduced a robust optimisation framework that accounts for both supply and demand uncertainties. They applied this framework to a multi-period, single-station inventory problem and extended its application to multi-echelon cases. The resulting robust policies offer cost benefits by effectively addressing concerns related to both supply and demand uncertainty. An approach to inventory problems using fuzzy logic was presented by Yao, Chang, and Su (2000), who proposed a fuzzy economic order quantity model with uncertainties about both the order quantity and the total demand quantified by triangular fuzzy numbers.

In this paper, we propose an alternative approach to inventory management with random demand. Instead of relying on a predetermined probability distribution for demand, we apply nonparametric predictive inference (NPI). NPI is a frequentist statistics method based on only few assumptions, leading to inferences which are strongly based on available data. This method can be used if one has little information about the random demand beyond the data, or if one does not want to use any further information or an assumed probability distribution for the random demand.

NPI and fuzzy methods both serve as means to take uncertainty into account, yet they employ different mathematical frameworks and underlying philosophies. NPI focuses on making predictions without assuming a specific distribution, while fuzzy methods handle uncertainty by incorporating the concept of partial membership in sets. The selection between these methods should be contingent on the specific characteristics of the inventory planning problem, the nature of the data involved, and the type of uncertainty under consideration. NPI is particularly created in an attempt to base decisions on the data with minimal further assumptions beyond the exchangeability of the random quantities, which for inventory scenarios can best be interpreted as absence of patterns in the random demands. No subjective assumptions, such as prior distributions (in Bayesian methods) or membership functions (in fuzzy logic) are added to the data in the NPI framework. Of course, this does imply that data on previous demands must be available in order to apply the NPI methods presented in this paper.

During the last two decades, NPI (Augustin & Coolen, 2004; Coolen, 2006) has been developed as a frequentist statistical framework based on the assumption $A_{(n)}$, proposed by Hill (1968, 1988), which provides direct probabilities for one or more future observations based on n observations of related random quantities. Inferences based on $A_{(n)}$ are nonparametric and predictive, and have also been called 'low structure inferences', since these inferences are based on limited assumptions (Geisser, 1993). The explicitly predictive nature of NPI makes it particularly useful for a range of topics in Operational Research. For example, NPI has been presented for queueing (Coolen & Coolen-Schrijner, 2003) and age replacement scenarios (Coolen-Schrijner, Shaw, & Coolen, 2009). NPI has been developed for a variety of data types, including Bernoulli data (Coolen, 1998) and right-censored data (Coolen & Yan, 2004). Recently, NPI has also been applied to finance problems (Chen, Coolen, & Coolen-Maturi, 2019; He, Coolen, & Coolen-Maturi, 2019).

This paper presents the first application of NPI to inventory theory. NPI is a powerful statistical methodology, in particular as it is the only nonparametric method which is exactly calibrated (Lawless & Fredette, 2005), which means that the resulting predictive inferences are valid within the frequentist statistics framework. It is important to emphasize that this property holds for any number of observations of previous demands. The setting of this paper may not be directly relevant to a wide range of practical inventory decision scenarios, which require further research, similar to the wide variety of research directions that were build on the first theoretical contributions to inventory theory, which were based on assumed probability distributions for the demands.

This paper is organised as follows. Section 2 provides brief reviews of the classical inventory model with a single period. Section 3 provides a brief introduction to NPI. Section 4 presents NPI for the single-period inventory model, with the aim to select an optimal inventory level using several optimality criteria. In Section 5, the NPI method and the classical method for a single-period model are compared through a simulation study is presented. Finally, in Section 6 some concluding remarks are provided.

2. Classical inventory model

More than hundred years ago, Harris (1913) presented a basic model to support inventory decisions. Many researchers have extended Harris's model, considering different types of demands and replenishment (Hadley & Whitin, 1963; Taha, 2017). Inventory models have been classified based on the pattern of demand, which may be either deterministic or random over time. Deterministic demand occurs when we know the exact quantities needed over a period of time. Random demand occurs when the demand is not known within a period of time. Forecasting the random demand tends to be a challenge for the decision maker (Van Hees & Monhemius, 1972). In most of the literature on inventory problems, this difficulty is avoided by the assumption of a fully specified stochastic model for the random demand.

2.1. Single-period inventory model

The single-period inventory problem considers the scenario in which an inventory is needed to satisfy demand during only a single period. This means that only once the level of inventory can be decided, and that demand cannot be satisfied in a later period (Collier & Evans, 2010). According to Reid and Sanders (2007), the single-period model is typically designed for goods with specific characteristics: they are sold at their regular price only during the single time period and the salvage value of goods left over at the end of the period is less than their original cost, so the inventory owner loses money when goods are sold for their salvage value. An example is a newsagent who orders newspapers for a specific day, where the problem for the newsagent is to decide how many newspapers to order (Hadley & Whitin, 1963; Kao & Hsu, 2002; Taha, 2017). In general, the demand, denoted by D, can be either a continuous or a discrete random quantity, in this paper we restrict attention to continuous demand. One can use a variety of optimality criteria for the inventory level, we will consider maximisation of the probability of non-negative profit and maximisation of the expected profit. We

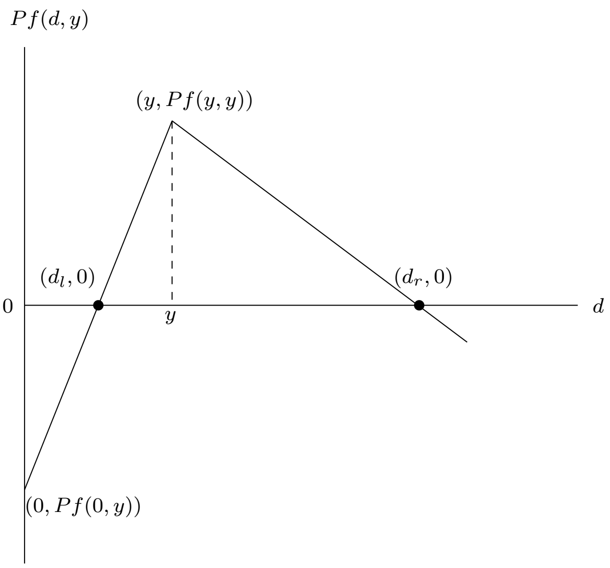

Figure 1. Profit for fixed inventory level y as function of demand d

assume that the inventory level is y, we aim at determining the best value of y, which we denote by y^*. The costs considered in the basic single-period inventory model are as follows:

(1) Purchase cost c is the purchase price per unit ordered, we assume $c > 0$. The total purchase costs are cy.
(2) Price p is the price per unit sold, we assume a selling price that is high enough to cover the costs of inventory. The total amount of money from sales is $p \min(D, y)$.
(3) Holding cost h is the cost per unit for unsold items remaining at the end of the period. The total holding costs are equal to $h(y-D)^+$, where $(u)^+ := \max(0, u)$. Note that h can be negative, reflecting the possibility of some income resulting from off-loading unsold items.
(4) Shortage cost s is the cost per unit of demand that cannot be met, we assume $s \geq 0$. The total shortage costs are $s(D-y)^+$.

According to Waters (2008), these costs lead to the profit function

$$Pf(D, y) = p \min(D, y) - cy - h(y-D)^+ - s(D-y)^+ \tag{1}$$

We consider two main optimality criteria in this paper, these are discussed next.

2.2. Maximisation of the probability of non-negative profit

One may wish to order in such a way that the probability of a loss is minimal, hence to maximise the probability of a non-negative profit (Sankarasubramanian & Kumaraswamy, 1983). This criterion is straightforwardly adapted to maximisation of the probability that the profit exceeds a specific value other than zero, we do not address this further.

For fixed inventory level y, the profit function, given by Equation (1), is a function of the demand $D = d$, and it consists of two line segments as presented in Figure 1.

From Equation (1), we find d_l and d_r such that $Pf(d_l, y) = Pf(d_r, y) = 0$. For $d < y$,

$$d_l = \frac{(h+c)y}{p+h} \tag{2}$$

and for $d > y$,

$$d_r = \frac{(p+s-c)y}{s} \tag{3}$$

Note that d_l and d_r are functions of y, we do not explicitly include this in the notation. The profit is greater than or equal to zero whenever $d_l \leq D \leq d_r$. So,

$$P(Pf(D, y) \geq 0) = P(d_l \leq D \leq d_r) = \int_{d_l}^{d_r} f_D(u)du \tag{4}$$

where $f_D(\cdot)$ is the probability density function (PDF) for the demand D. Let y^*_{CP} denote the optimal inventory level which maximises the probability that the profit is greater than or equal to zero. Setting the first derivative with respect to y of this probability equal to zero leads to

$$\frac{p+s-c}{s} f_D(d_r) - \frac{h+c}{p+h} f_D(d_l) = 0 \tag{5}$$

For any given probability distribution for D, Equation (5) provides the possible value(s) for y^*_{CP}, checking the second-order sufficient condition for optimality leads to the optimal inventory level.

2.3. Maximisation of the expected profit

The expected profit for inventory level y is (Shih, 1979)

$$E(Pf(D, y)) = \int_0^y (pu - h(y-u) - cy) f_D(u)du + \int_y^\infty (py - s(u-y) - cy) f_D(u)du \tag{6}$$

The optimal inventory level y^*_{CE}, which maximises the expected profit, is derived by setting the first derivative of this expected profit to zero, leading to the equation

$$P(D \leq y^*_{CE}) = \frac{p+s-c}{p+s+h} \tag{7}$$

As the second derivative of the expected profit is negative at all values of y with $f_D(y) > 0$, the value y^*_{CE} resulting from Equation (7) is the optimal inventory level when aiming at maximisation of the expected profit.

3. Nonparametric predictive inference

Nonparametric Predictive Inference (NPI) is a frequentist statistical framework providing probabilities for future observations. NPI is based on the assumption $A_{(n)}$ proposed by Hill (1968, 1988), which is described as follows. Consider $n+1$ real-valued exchangeable random quantities $X_1, X_2, \ldots, X_n, X_{n+1}$. Let the ordered observations of $X_1, \ldots, X_n$ be denoted by $x_1 < x_2 < \ldots < x_n$. Note that, throughout this paper, we assume that ties do not occur, for ease of presentation; if there are ties in data then these can easily be broken by increasing an observation by a neglectably small amount. Also for ease of notation, let $x_0 = -\infty$ and $x_{n+1} = \infty$, (these can be replaced by other known bounds of the support of the X_i). The n observations partition the real-line into $n+1$ intervals, $I_j = (x_{j-1}, x_j)$ for $j = 1, \ldots, n+1$. The assumption $A_{(n)}$ is:

$$P(X_{n+1} \in I_j) = \frac{1}{n+1} \text{ for all } j = 1, \ldots, n+1 \tag{8}$$

Without any further assumptions on the distribution of the probability masses within the intervals I_j, $A_{(n)}$ is not sufficient to provide precise probabilities for many events of interest involving X_{n+1}. However, optimal bounds for probabilities for such events, corresponding to $A_{(n)}$, can be derived using de Finetti's fundamental theorem of probability (De Finetti, 1974). These bounds are lower and upper probabilities in the theory of imprecise probability (Augustin & Coolen, 2004; Augustin, Coolen, de Cooman, & Troffaes, 2014; Walley, 1991). The NPI lower and upper probabilities for the event $X_{n+1} \in B$, with $B \subset \mathbb{R}$, are the maximum lower bound and the minimum upper bound, respectively, for the probability for this event, based only on the $A_{(n)}$ assumption (Augustin & Coolen, 2004). The NPI lower probability is

$$\underline{P}(X_{n+1} \in B) = \sum_{j=1}^{n+1} 1\{I_j \subseteq B\} P(X_{n+1} \in I_j) \tag{9}$$

and the corresponding NPI upper probability is

$$\overline{P}(X_{n+1} \in B) = \sum_{j=1}^{n+1} 1\{I_j \cap B \neq \emptyset\} P(X_{n+1} \in I_j) \tag{10}$$

where $1\{A\}$ is the indicator function, which is equal to 1 if event A occurs and 0 else. The NPI lower probability in Equation (9) is obtained by summing only the probability masses, resulting from the assumption $A_{(n)}$, that are necessarily assigned to B, while the NPI upper probability in Equation (10) is derived by summing all probability masses that can be assigned to B.

Due to the absence of any distributional assumptions, the results of applying NPI do not generally coincide with results from classical or Bayesian statistics based on assumed probability distributions. However, the NPI inferences are related to the use of the empirical distributions based on data, as the empirical cumulative distribution function based on data $x_1 < x_2 < \ldots < x_n$ is bounded by the corresponding NPI lower and upper cumulative distribution functions. Compared to the possible use of the empirical distribution function, the NPI approach has the advantages that it is explicitly in terms of a future observation, which is suitable for inventory problems, and that the imprecision, which is the difference between corresponding NPI upper and lower probabilities, reflects the amount of data on which the inferences are based.

It is important to emphasize that the assumption $A_{(n)}$, given in Equation (8), does not make any assumptions about the spread of the probability mass $1/(n+1)$ within each interval I_j. To develop the NPI approach to inventory decision problems, we need an additional concept and notation for probability mass assigned to an interval without further assumptions on the spread of the probability mass within the interval. We use the M-function (Coolen & Yan, 2004), which is introduced as follows. The probability mass for a random quantity X assigned to an interval (a, b) can be denoted by $M_X(a, b)$ and referred to as M-function value for X on (a, b). We may remove the subscript X and just use $M(a, b)$ if it is clear which random quantity is involved. The concept of M-function is similar to that of Shafer's basic probability assignments (Shafer, 1976). Clearly, each M-function value should be in $[0, 1]$ and all M-function values for a random quantity must sum to one. The important difference of M-function values compared to the standard specification of probabilities is that the intervals which have been assigned positive probability mass, according to the M-function specification, do not need to be disjoint. In the next section the derivation of the M-function values for use in the single-period inventory problem is explained.

4. NPI for the single-period inventory model

In this section, we develop NPI for decision support in the single-period inventory model. We assume that data of demand in n previous periods are available, ordered as $d_1 < d_2 < \ldots < d_n$, and we consider the random demand D_{n+1} for the next single period for which the inventory decision is required. We assume that there is a known upper bound for the demand, denoted by d_u, which is logically greater than d_n, and that demand is positive. Based on these data, and setting $d_0 = 0$ and $d_{n+1} = d_u$, the assumption $A_{(n)}$ for D_{n+1} leads to:

$$P(D_{n+1} \in (d_{j-1}, d_j)) = \frac{1}{n+1} \text{ for all } j = 1, \ldots, n+1 \tag{11}$$

The essential step in developing NPI for the single-period inventory model, is the transfer of the partial probability distribution specification for D_{n+1} to a partial probability distribution specification for the profit, using the profit function given by Equation (1). This process is illustrated in Figure 2, and involves the profit $Pf(D_{n+1}, y)$ as function of the random demand D_{n+1} and fixed inventory level y. The overall aim is to determine an optimal value for y. Let j_y be such that $y \in (d_{j_y-1}, d_{j_y})$. The probabilities for D_{n+1}, given in Equation (11), lead to the following M-function values for the random profit $Pf(D_{n+1}, y)$,

$$M(Pf(d_{j-1}, y), Pf(d_j, y)) = \frac{1}{n+1} \quad \text{for } j \in \{1, \ldots, j_y - 1\} \tag{12}$$

$$M(\min[Pf(d_{j_y-1}, y), Pf(d_{j_y}, y)], Pf(y, y)) = \frac{1}{n+1} \quad \text{for } j = j_y \tag{13}$$

$$M(Pf(d_j, y), Pf(d_{j-1}, y)) = \frac{1}{n+1} \quad \text{for } j \in \{j_y + 1, \ldots, n+1\} \tag{14}$$

The transfer of the probabilities for D_{n+1}, as given in Equation (11), to M-function

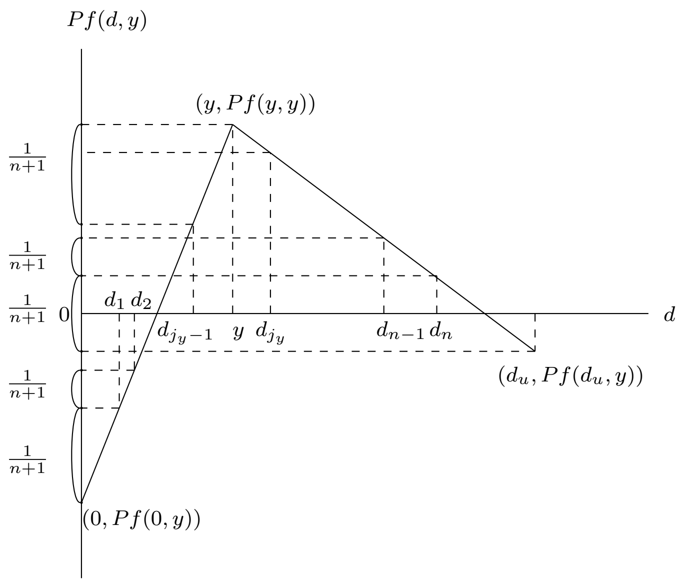

Figure 2. Single-period inventory - M-functions

values for $Pf(D_{n+1}, y)$ for the cases in Equations (12) and (14) follow straightforwardly from Figure 2. For intervals of possible values d for the future demand D_{n+1} with $d < y$, the probability $1/(n+1)$ for the event that the future demand D_{n+1} is in a specific interval (d_{j-1}, d_j) logically translate to the probability for the corresponding profit $Pf(D_{n+1}, y)$ to be in the interval $(Pf(d_{j-1}, y), Pf(d_j, y))$ with the same probability $1/(n+1)$, because $d_j < y$ and the profit function $Pf(d, y)$ is increasing as function of d for $d < y$. This is illustrated in Figure 2 and leads to Equation (12). The derivation of the M-function values in Equation (14), for intervals of the demand value greater than y, is similar, with the difference that the profit function in this range is decreasing as function of d.

The derivation of the M-function value given in Equation (13) is also best understood from Figure 2, where we should emphasize once more that no assumptions are made about the probability mass $\frac{1}{n+1}$ for D_{n+1} assigned to the interval (d_{j_y-1}, d_{j_y}). Therefore, the probability mass $1/(n+1)$ for the event $D_{n+1} \in (d_{j_y-1}, d_{j_y})$ leads to this same probability mass being assigned to the event that the profit $Pf(D_{n+1}, y)$ is in the interval $(\min[Pf(d_{j_y-1}, y), Pf(d_{j_y}, y)], Pf(y, y))$. Note here that the intervals to which probability masses $1/(n+1)$ are assigned for the profit $Pf(D_{n+1}, y)$, according to Equations (12)-(14), overlap, so to develop NPI for inventory problems the concept of M-function is required, as briefly explained in the previous section, as such a specification cannot be achieved by the classical probability concept which requires complete and unique specification of probability for each subset of the real-values in case of a real-valued random quantity.

Subsection 4.1 presents the NPI lower and upper probabilities for the event that the profit over the single period considered is non-negative, and uses these to determine an optimal inventory level. This is followed by focus on the NPI lower and upper expected values for the profit, and their optimisation, in Subsection 4.2.

4.1. NPI lower and upper probabilities for non-negative profit

When aiming to maximise the NPI lower probability for non-negative profit, it is easily understood from Figures 1 and 2 that we should maximise the number of intervals (d_{j-1}, d_j), for $j = 1, \ldots, n+1$, that are entirely within $[d_l, d_r]$, where we note that d_l and d_r are functions of the inventory level y. A useful insight is that the optimisation leading to the NPI lower probability can be performed by only considering cases with $d_l = d_j$, for $j = 1, \ldots, n+1$, because for any d_l in (d_{j-1}, d_j) we can instead move d_l to be equal to d_j without reducing the number of intervals, based on the data, that must be within $[d_l, d_r]$, whilst possibly increasing this number because the d_r that corresponds to the d_l will also increase when d_l increases. This is the main idea behind the following method for maximising the NPI lower probability for non-negative profit.

Let y_1 be such that $Pf(d_1, y_1) = 0$ with $y_1 \geq d_1$. Hence, y_1 is such that the increasing line segment of the function $Pf(D, y_1)$ is equal to 0 at $D = d_1$, so related to Figure 1 we have $d_l = d_1$. This leads to

$$y_1 = \frac{(p+h)d_1}{c+h}$$

This value y_1 fully specifies the profit function for given values of the costs. Let d_1^r be such that $Pf(d_1^r, y_1) = 0$ and $d_1^r \geq y_1$, hence the decreasing line segment of the function $Pf(D, y_1)$ is equal to 0 at $D = d_1^r$. By Equations (2) and (3) we have

$$d_1^r = \frac{(p+s-c)y_1}{s} = \frac{(p+s-c)(p+h)d_1}{s(c+h)}$$

The NPI lower probability $\underline{P}(Pf(D_{n+1}, y_1) \geq 0)$ is derived by counting the number of intervals (d_{j-1}, d_j) which are entirely contained in $[d_1, d_1^r]$, noting that each such interval has been assigned probability mass $\frac{1}{n+1}$ based on the $A_{(n)}$ assumption for the future demand D_{n+1}. Let n_1 denote the number of d_j, for $j = 1, \ldots, n+1$, such that $d_1 \leq d_j \leq d_1^r$, so to summarize, we have

$$y_1 = \frac{(p+h)d_1}{c+h} \tag{15}$$

$$d_1^r = \frac{(p+s-c)(p+h)d_1}{s(c+h)} \tag{16}$$

$$n_1 = \#\{d_j \ : \ d_j \in [d_1, d_1^r], j \in \{1, \ldots, n+1\}\} \tag{17}$$

There are $(n_1 - 1)^+$ intervals (d_{j-1}, d_j) which are entirely in $[d_1, d_1^r]$, so the NPI lower probability for the event $Pf(D_{n+1}, y_1) \geq 0$ is

$$\underline{P}(Pf(D_{n+1}, y_1) \geq 0) = \frac{n_1 - 1}{n+1} \tag{18}$$

We now perform the same steps as above with $d_l = d_k$, for $k = 2, \ldots, n$. The profit function has value 0 on the increasing line segment at d_k. The corresponding y_k, the value d_k^r where the profit function has value 0 on its decreasing line segment, and the number n_k of d_j in $[d_k, d_k^r]$, are

$$y_k = \frac{(p+h)d_k}{c+h} \tag{19}$$

$$d_k^r = \frac{(p+s-c)y_k}{s} \tag{20}$$

$$n_k = \#\{d_j \ : \ d_j \in [d_k, d_k^r], j \in \{1, \ldots, n+1\}\} \tag{21}$$

This leads to

$$\underline{P}(Pf(D_{n+1}, y_k) \geq 0) = \frac{n_k - 1}{n+1} \tag{22}$$

An optimal inventory level $y^*_{\underline{P}}$ which maximises $\underline{P}(Pf(D_{n+1}, y))$ is now derived by setting $y^*_{\underline{P}} = y_k$, with $k \in \{1, \ldots, n\}$ such that the corresponding value of n_k is maximal. Note that it is quite likely that there is not a unique inventory level that maximises $\underline{P}(Pf(D_{n+1}, y))$, not only because there may not be a unique value n_k leading to the maximum $\underline{P}(Pf(D_{n+1}, y))$, but also because values just less than $y^*_{\underline{P}}$ are likely to lead to the same value for this NPI lower probability for non-negative profit.

Deriving an optimal inventory level to maximise the NPI upper probability for the event $Pf(D_{n+1}, y) \geq 0$ can be done in a similar way as for the corresponding NPI lower probability. This upper probability is derived by counting the intervals (d_{j-1}, d_j) that have a non-empty intersection with $[d_l, d_r]$, which is again easily seen from Figures 1 and 2. Of course, this includes all the intervals that were included in the counts to derive the NPI lower probability, while some attention is needed for the intervals which contain either d_l or d_r. Assuming that $D_{n+1} = 0$ leads to negative profit, which is the case for any positive inventory level, then the interval (d_{j-1}, d_j) containing d_l must now be included in the count.

For the interval which contains d_r, we need to consider the value of the profit, for given y, at the end-point d_u of the support for the demand. If d_u is large enough to have a negative profit at d_u for given y, then the interval (d_{j-1}, d_j) containing d_r must now be included in the count, but if the profit at d_u is non-negative, then that interval will already have been included in the count for the NPI lower probability. This leads to the following NPI upper probability for inventory levels y_k, for $k = 1, \ldots, n$, as defined in Equations (15) and (19),

$$\overline{P}(Pf(D_{n+1}, y_k) \geq 0) = \begin{cases} \underline{P}(Pf(D_{n+1}, y_k) \geq 0) + \frac{2}{n+1} & \text{if } Pf(d_u, y_k) < 0 \\ \underline{P}(Pf(D_{n+1}, y_k) \geq 0) + \frac{1}{n+1} & \text{if } Pf(d_u, y_k) \geq 0 \end{cases} \tag{23}$$

The optimal inventory level $y^*_{\overline{P}}$ which maximises $\overline{P}(Pf(D_{n+1}, y) \geq 0)$ is easily derived by taking the value y_k which maximises this upper probability over $k = 1, \ldots, n$. The same comment with regard to non-uniqueness of the optimal inventory level, as made above for the NPI lower probability, applies here. Furthermore, we should mention that, in this section, we have taken some liberties on mathematical accuracy for the sake of simplicity of the presentation: First, for the optimisation of the NPI upper probability, the arguments provided would actually require consideration of values $d_k - \epsilon$ instead of d_k, for very small $\epsilon > 0$, but the effect on the corresponding strategies y_k, and hence on the optimal inventory level, is neglectable. Secondly, we have not paid attention to situations where d_l or d_u could coincide with an observed value d_j. This is of no serious consequence when demand is a continuous random quantity; if one would want to model discrete demand then more care would be needed, but one would then also want to consider a different NPI approach than the method for continuous random quantities used in this paper; this is not considered further here.

As optimisation of the NPI lower probability and the NPI upper probability for non-negative profit may not both lead to the same optimal inventory level, one may need to choose which of these two criteria to use. One could consider optimisation of the NPI lower probability a somewhat pessimistic perspective, with optimisation of the NPI upper probability reflecting a more optimistic point of view. However, one could also combine these two criteria using the Hurwicz criterion (Arrow & Hurwicz, 1977), which here implies maximisation of a weighted average of the NPI lower and upper probabilities. With $\omega \in [0,1]$, we define

$$H_{P,\omega}(Pf(D_{n+1},y_k) \geq 0) = \omega \underline{P}(Pf(D_{n+1},y_k) \geq 0) + (1-\omega)\overline{P}(Pf(D_{n+1},y_k) \geq 0) \tag{24}$$

The parameter ω can be interpreted as an optimism-pessimism index. The choice of ω, like the overall choice of the optimality criterion, would be for the decision maker to make. Using Equations (22) and (23) we get

$$H_{P,\omega}(Pf(D_{n+1},y_k) \geq 0) = \begin{cases} \dfrac{(n_k+1-2\omega)^+}{n+1} & \text{if } Pf(d_u,y_k) < 0 \\ \dfrac{(n_k-\omega)^+}{n+1} & \text{if } Pf(d_u,y_k) \geq 0 \end{cases} \tag{25}$$

The optimal inventory level according to the Hurwicz criterion will be denoted by $y^*_{P,\omega}$. Example 4.1 illustrates the methods presented in this section, considering the optimisation of the NPI lower probability, the NPI upper probability, and the corresponding Hurwicz criterion.

Example 4.1. Consider an inventory system with the following costs: $p = 50$, $c = 20$, $h = 10$, $s = 20$, and assume that demand is known to be between $d_0 = 0$ and $d_u = 40$. Assume that there are $n = 5$ demand observations, with values $7.2, 12.5, 15.3, 22.6, 35.4$. Table 1 presents the results of the method presented in this section, specifying the NPI lower and upper probabilities for the event that the profit for the single period considered will be non-negative, together with the corresponding values of y_k and of the quantities d^r_k and n_k that are part of the computations as explained above. The NPI lower probability is maximal at $y_1 = 14.4$ and $y_2 = 25$, so either of these values can be chosen as $y^*_{\underline{P}}$. The NPI upper probability is maximal at $y^*_{\overline{P}} = y_1 = 14.4$. Maximisation of the Hurwicz criterion leads to $y^*_{P,\omega} = 14.4$ for all $\omega \in [0,1)$, while for $\omega = 1$ this criterion is the same as maximisation of the NPI lower probability, which was discussed above.

4.2. NPI lower and upper expected profits

In this section, we present the NPI lower and upper expected profits for the next period, as function of the inventory level y. The derivations are based on the M-functions illustrated in Figure 2 and presented in Equations (12)-(14).

Table 1. NPI lower and upper probabilities for Example 4.1

k	y_k	d_k^r	n_k	$\underline{P}(Pf(D_6, y_k) \geq 0)$	$\overline{P}(Pf(D_6, y_k) \geq 0)$
1	14.4	36.0	5	0.67	1.00
2	25.0	62.5	5	0.67	0.83
3	30.6	76.5	4	0.50	0.67
4	45.2	113.0	3	0.33	0.50
5	70.8	177.0	2	0.17	0.33

The NPI lower expected profit, denoted by $\underline{E}_{n+1}(y)$, is derived by assigning the probability masses $\frac{1}{n+1}$, according to the M-function values in Equations (12)-(14), to the minimum (or infimum) value for $Pf(D_{n+1}, y)$ per interval. With, as before, j_y such that $y \in (d_{j_y-1}, d_{j_y})$, this leads to

$$\begin{aligned}
\underline{E}_{n+1}(y) &= \sum_{j=1}^{j_y-1} M(Pf(d_{j-1}, y), Pf(d_j, y))Pf(d_{j-1}, y) \\
&\quad + M(\min[Pf(d_{j_y-1}, y), Pf(d_{j_y}, y)], Pf(y, y)) \min[Pf(d_{j_y-1}, y), Pf(d_{j_y}, y)] \\
&\quad + \sum_{j=j_y+1}^{n+1} M(Pf(d_j, y), Pf(d_{j-1}, y))Pf(d_j, y) \\
&= \frac{1}{n+1}\left(\sum_{j=1}^{j_y-1} Pf(d_{j-1}, y) + \min[Pf(d_{j_y-1}, y), Pf(d_{j_y}, y)] + \sum_{j=j_y+1}^{n+1} Pf(d_j, y) \right)
\end{aligned}$$

$$\begin{aligned}
&= \frac{1}{n+1}\Bigg((j_y - 1)(-(c+h)y) + (p+h) \sum_{j=1}^{j_y-1} d_{j-1} \\
&\quad + \min[-(c+h)y + (p+h)d_{j_y-1}, (p-c+s)y - sd_{j_y}] \\
&\quad + (n+1-j_y)(p-c+s)y - s \sum_{j=j_y+1}^{n+1} d_j \Bigg)
\end{aligned} \tag{26}$$

To determine an optimal inventory level, $y^*_{\underline{E}}$, which maximises $\underline{E}_{n+1}(y)$, we introduce the following two functions,

$$\begin{aligned}
\underline{E}_a(y) &= \frac{1}{n+1}\Bigg((j_y - 1)(-(c+h)y) + (p+h) \sum_{j=1}^{j_y-1} d_{j-1} - (c+h)y + (p+h)d_{j_y-1} \\
&\quad + (n+1-j_y)(p-c+s)y - s \sum_{j=j_y+1}^{n+1} d_j \Bigg)
\end{aligned} \tag{27}$$

and

$$\underline{E}_b(y) = \frac{1}{n+1}\Bigg((j_y - 1)(-(c+h)y) + (p+h)\sum_{j=1}^{j_y-1} d_{j-1} + (p-c+s)y - sd_{j_y}$$
$$+ (n+1-j_y)(p-c+s)y - s\sum_{j=j_y+1}^{n+1} d_j\Bigg) \tag{28}$$

By Equation (26), $\underline{E}_{n+1}(y) = \min[\underline{E}_a(y), \underline{E}_b(y)]$. It is easy to verify that the function $\underline{E}_{n+1}$ discontinuous at d_l, for all $l \in \{1, \ldots, n\}$. $\underline{E}_a(y)$ and $\underline{E}_b(y)$ are linear functions in each interval $[d_{j_y-1}, d_{j_y}]$. $\underline{E}_a(y)$ is increasing in $[d_{j_y-1}, d_{j_y}]$ if and only if

$$j_y < \frac{(n+1)(p-c+s)}{p+h+s} =: K_1 \tag{29}$$

and $\underline{E}_a(y)$ is decreasing in $[d_{j_y-1}, d_{j_y}]$ if and only if $j_y > K_1$. $\underline{E}_b(y)$ is increasing in $[d_{j_y-1}, d_{j_y}]$ if and only if

$$j_y < \frac{c+h+(n+2)(p-c+s)}{p+h+s} = K_1 + 1 \tag{30}$$

and $\underline{E}_b(y)$ is decreasing in $[d_{j_y-1}, d_{j_y}]$ if and only if $j_y > K_1 + 1$. This implies that the maximum value of $\underline{E}_{n+1}(y) = \min[\underline{E}_a(y), \underline{E}_b(y)]$ is at the intersection point of $\underline{E}_a(y)$ and $\underline{E}_b(y)$ in the single interval where $\underline{E}_a(y)$ decreases and $\underline{E}_b(y)$ increases. This leads to optimal inventory level, which maximises the NPI lower expected profit,

$$y^*_{\underline{E}} = \frac{(p+h)d_{j_y-1} + sd_{j_y}}{p+h+s} \tag{31}$$

where $K_1 \le j_y < K_1 + 1$. This is illustrated in the following example.

Example 4.2. Let the costs be $p = 103$, $c = 16$, $h = 20$ and $s = 7$, and suppose data consisting of $n = 9$ observations: $2.2, 3.7, 5.4, 7.7, 10.1, 12.6, 15.2, 17.9, 20.7$. We assume that the maximum possible value for the demand is $d_u = 22.9$.

We have $K_1 = 7.23$, so, in the first seven intervals of the partition of $[0, 22.9]$ created by the data, $\underline{E}_a(y)$ is increasing and thereafter it is decreasing, while $\underline{E}_b(y)$ is increasing in the first eight intervals, and decreasing thereafter. This is illustrated in Figure 3. So the minimum of these two functions reaches its maximum value at the intersection of these two functions in the interval (d_7, d_8), leading to so $y^*_{\underline{E}} = 15.35$ with corresponding NPI lower expected profit $\underline{E}_{n+1}(y^*_{\underline{E}}) = 515.90$. $\Diamond$

Similarly, by assigning the probability masses $\frac{1}{n+1}$ in Equations (12)-(14) to the

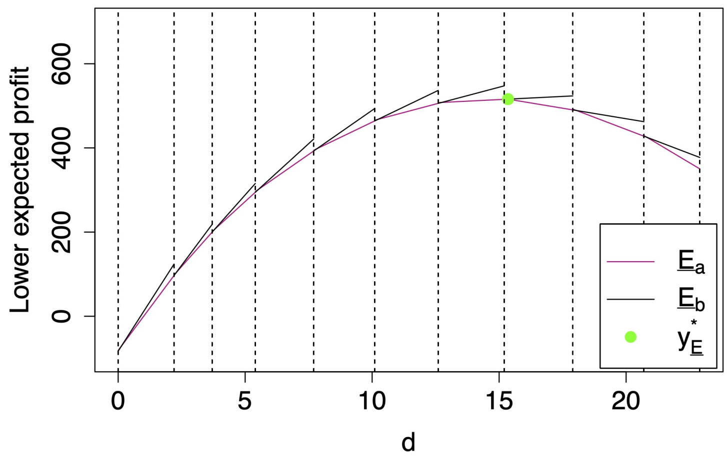

Figure 3. The functions $\underline{E}_a(y)$ and $\underline{E}_b(y)$ for Example 4.2

maximal value for $Pf(D_{n+1}, y)$ per interval, we derive the upper expected profit

$$\begin{aligned}
\overline{E}_{n+1}(y) &= \sum_{j=1}^{j_y-1} M(Pf(d_{j-1}, y), Pf(d_j, y))Pf(d_j, y) \\
&\quad + M(\min[Pf(d_{j_y-1}, y), Pf(d_{j_y}, y)], Pf(y, y))Pf(y, y) \\
&\quad + \sum_{j=j_y+1}^{n+1} M(Pf(d_j, y), Pf(d_{j-1}, y))Pf(d_{j-1}, y) \\
&= \frac{1}{n+1}\left(\sum_{j=1}^{j_y-1} Pf(d_j, y) + Pf(y, y) + \sum_{j=j_y+1}^{n+1} Pf(d_{j-1}, y)\right) \\
&= \frac{1}{n+1}\Bigg((j_y - 1)(-(c+h)y) + (p+h)\sum_{j=1}^{j_y-1} d_j + (p-c)y \\
&\quad + (n+1-j_y)(p-c+s)y - s\sum_{j=j_y+1}^{n+1} d_{j-1}\Bigg) \qquad (32)
\end{aligned}$$

It is easy to show that $\overline{E}_{n+1}(y)$ is a continuous function. To derive the optimal inventory level which maximises $\overline{E}_{n+1}(y)$, which we denote by $y^*_{\overline{E}}$, we use that $\overline{E}_{n+1}(y)$ is increasing over the interval $[d_{j_y-1}, d_{j_y}]$ if and only if

$$j_y < \frac{h + p + (n+1)(p-c+s)}{p+s+h} =: K_2 \qquad (33)$$

and $\overline{E}_{n+1}(y)$ is decreasing over the interval $[d_{j_y-1}, d_{j_y}]$ if and only if $j_y > K_2$. This implies that $y^*_{\overline{E}} = d_{l^*}$, with l^* the largest value in $\{1, 2, \ldots, n\}$ which is less than K_2. This is illustrated in the following example.

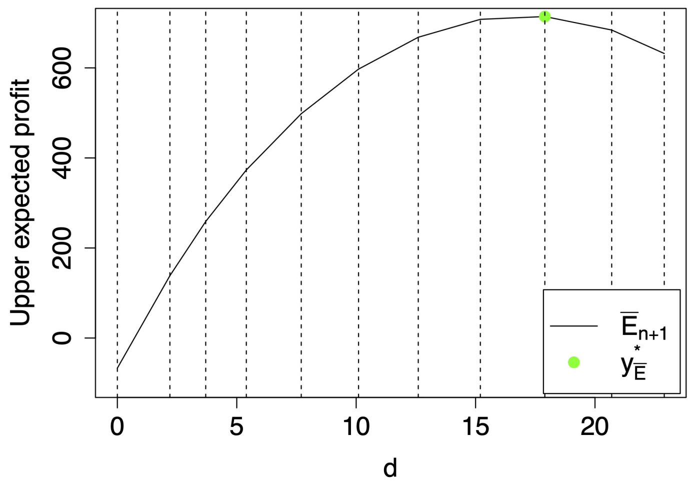

Figure 4. The NPI upper expected profit function $\overline{E}_{n+1}(y)$

Example 4.3. Consider the same scenario as in Example 4.2, the aim is to find the optimal inventory level which maximises the NPI upper expected profit. The upper expected profit function $\overline{E}_{n+1}(y)$ is presented in Figure 4. We have $K_2 = 8.18$, so $y^*_{\overline{E}} = d_8$, that is the 8th ranked observation out of the 9 data observations, hence $y^*_{\overline{E}} = 17.9$ with corresponding NPI upper expected profit $\overline{E}_{n+1}(y^*_{\overline{E}}) = 714.02$. $\diamondsuit$

It is worth noting that, in Examples 4.2-4.3, the optimal inventory level achieved by maximising the NPI upper expected profit is greater than for the NPI lower expected profit. This is intuitively logical since the NPI upper expected profit is based on an optimistic view for the demand in the next period, while the NPI lower expected profit is based on a pessimistic view. Hence, one would likely end up with an optimal inventory level that is higher under the optimistic view than under the pessimistic view. A general result along these lines has not been achieved and is left as an interesting challenge for future research.

As an alternative to maximising the NPI lower expected profit or the NPI upper expected profit, one can use the Hurwicz criterion and aim at maximising their weighted average,

$$H_{E,\omega}(y) = \omega \underline{E}_{n+1}(y) + (1-\omega)\overline{E}_{n+1}(y) \tag{34}$$

for $0 < \omega < 1$. Using Equations (26) and (32) leads to

$$
\begin{aligned}
H_{E,\omega}(y) = \frac{\omega}{n+1}\Big(& -(p-c)y + (p+h)(d_0 - d_{j_y-1}) - s(d_{n+1} - d_{j_y}) \\
& + \min[-(c+h)y + (p+h)d_{j_y-1}, (p-c+s)y - sd_{j_y}]\Big) \\
& + \frac{1}{n+1}\Big(y[(j_y-1)(-(c+h)) + (p-c) + (n+1-j_y)(p-c+s)] \\
& + (p+h)\sum_{j=1}^{j_y-1} d_j - s\sum_{j=j_y+1}^{n+1} d_{j-1}\Big)
\end{aligned} \tag{35}
$$

To determine an optimal inventory level, $y^*_{E,\omega}$, which maximises $H_{E,\omega}(y)$, we introduce the following two functions:

$$
\begin{aligned}
H_{\omega a}(y) = \frac{\omega}{n+1}\Big(& -(p-c)y + (p+h)(d_0 - d_{j_y-1}) - s(d_{n+1} - d_{j_y}) - (c+h)y \\
& + (p+h)d_{j_y-1}\Big) + \frac{1}{n+1}\Big(y[(j_y-1)(-(c+h)) + (p-c) \\
& + (n+1-j_y)(p-c+s)] + (p+h)\sum_{j=1}^{j_y-1} d_j - s\sum_{j=j_y+1}^{n+1} d_{j-1}\Big)
\end{aligned} \tag{36}
$$

and

$$
\begin{aligned}
H_{\omega b}(y) = \frac{\omega}{n+1}\Big(& -(p-c)y + (p+h)(d_0 - d_{j_y-1}) - s(d_{n+1} - d_{j_y}) + (p-c+s)y - sd_{j_y}\Big) \\
& + \frac{1}{n+1}\Big(y[(j_y-1)(-(c+h)) + (p-c) + (n+1-j_y)(p-c+s)] \\
& + (p+h)\sum_{j=1}^{j_y-1} d_j - s\sum_{j=j_y+1}^{n+1} d_{j-1}\Big)
\end{aligned} \tag{37}
$$

so $H_{E,\omega}(y) = \min[H_{\omega a}(y), H_{\omega b}(y)]$. The analysis to derive the optimal inventory level $y^*_{E,\omega}$ is similar to that for optimising the NPI lower expected profit, with $H_{E,\omega}$ also discontinuous at d_l, for all $l \in \{1, \dots, n\}$, and $H_{\omega a}(y)$ and $H_{\omega b}(y)$ linear functions in each interval $[d_{j_y-1}, d_{j_y}]$. $H_{\omega a}(y)$ is increasing in the interval $[d_{j_y-1}, d_{j_y}]$ if and only if

$$
j_y < \frac{(1-\omega)(p+h) + (n+1)(p-c+s)}{p+h+s} =: K_3 \tag{38}
$$

and $H_{\omega a}(y)$ is decreasing in the interval $[d_{j_y-1}, d_{j_y}]$ if and only if $j_y > K_3$. Similarly, $H_{\omega b}(y)$ is increasing in the interval $[d_{j_y-1}, d_{j_y}]$ if and only if

$$
j_y < \frac{\omega s + p + h + (n+1)(p-c+s)}{p+h+s} = K_3 + \omega \tag{39}
$$

and $H_{\omega b}(y)$ is decreasing in the interval $[d_{j_y-1}, d_{j_y}]$ if and only if $j_y > K_3 + \omega$. This

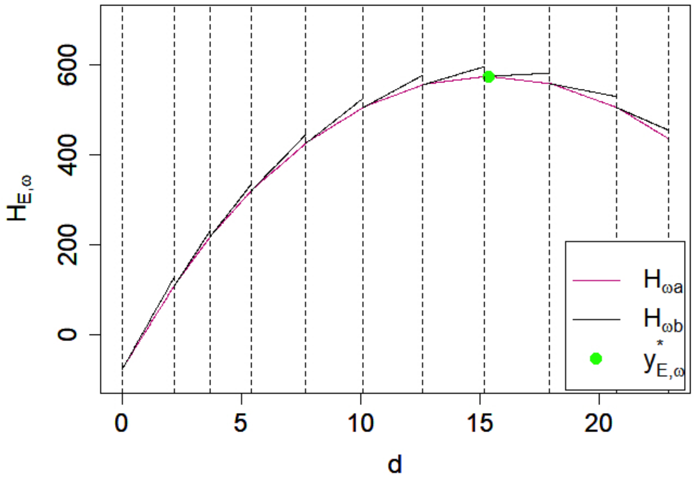

Figure 5. The functions $H_{\omega a}(y)$ and $H_{\omega b}(y)$ for Example 4.4

leads to optimal inventory level

$$y^*_{E,\omega} = \frac{(p+h)d_{j_2-1} + sd_{j_2}}{p+h+s} \tag{40}$$

where j_2 is the largest value in $\{1,\ldots,n\}$ for which $j_2 < K_3 + \omega$. This is illustrated in the following example.

Example 4.4. Consider again the same data as in Examples 4.2 and 4.3, and let $\omega = 0.7$. The aim is to find the optimal inventory level which maximises $H_{E,0.7}(y)$. We have $K_3 = 7.51$ so $j_2 = 8$ and the maximum value of $H_{E,0.7}(y)$ is achieved at the intersection of $H_{\omega a}(y)$ and $H_{\omega b}(y)$ in the interval (d_7, d_8), leading to $y^*_{E,0.7} = 15.35$ with corresponding $H_{E,0.7}(y^*_{E,0.7}) = 573.57$. This is illustrated in Figure 5. $\Diamond$

5. Simulation

This section presents the results of simulations to investigate the performance of the NPI methods for the single-period inventory problem, introduced in this paper. In each simulation run, $n+1$ values are generated from a Gamma distribution. The first n of these values are considered to be the data observations and used to determine the optimal inventory level y^* corresponding to one of the optimality criteria considered in Section 4. The remaining simulated value is considered to be the future observation, d_{n+1}, allowing the realised value of the profit function to be computed for the values of y^* and d_{n+1}.

We consider six different cases with regard to discrepancy between the model used for the data simulations, and the model assumed for the classical method to determine the optimal inventory level, which is compared to the optimal NPI inventory level. Each case is run 1,000 times and we report the number of these runs in which the profit resulting from the NPI method is greater than the profit resulting from the classical method. For all cases, we consider three different sample sizes, $n = 5, 50, 100$. The costs used are $c = 20$, $p = 50$, $h = 10$, $s = 20$ and we use $\omega = 0.5$ for the Hurwicz criterion. As finite end-point for the support of the random demand we use $d_u = 15$; in the rare event that a simulated value in a run exceeds 15 we delete the value and draw a new one; this has no real impact on the methods as the probability to get a value which exceeds 15 is very small for all models considered.

The Gamma(a, b) distribution, with shape parameter a and scale parameter b, has probability density function $f(x) = \frac{1}{\Gamma(a)b^a} x^{a-1} e^{\frac{-x}{b}}$ and mean value ab, and the Exponential(λ) distribution with rate λ has probability density function $f(x) = \lambda e^{-\lambda x}$ and mean value $\frac{1}{\lambda}$, both for $x > 0$. The cases considered are as follows, with first the model used for simulating the demands D specified, followed by the model assumed for the analysis according to the classical method. For the cases where the Gamma scale parameter b is simulated from the Uniform$(0, 2)$ distribution, one value is drawn and used for each run, so n observations are drawn using one specific value of b, and a new value of b is drawn for the next run.

- **Case I:** $D \sim$ Gamma$(3, 1)$, classical method: Gamma$(3, 1)$.
- **Case II:** $D \sim$ Gamma$(3, 1)$, classical method: Exp$(1/3)$.
- **Case III:** $D \sim$ Gamma$(3, 1)$, classical method: Exp$(1/2)$.
- **Case IV:** $D \sim$ Gamma$(3, b)$ with $b \sim$ Unif$(0, 2)$, classical method: Gamma$(3, 1)$.
- **Case V:** $D \sim$ Gamma$(3, 1)$, classical method: Exp(1).
- **Case VI:** $D \sim$ Gamma$(3, 1)$, classical method: Exp(2).

Case I is the scenario where the model used for the classical analysis is exactly the same as the model used for the data generation. In Case II the model for the analysis is Exponential but with the same mean value as the Gamma distribution used to generate the data. The further cases have other discrepancies between these two models, set up in such a way that we expected that the classical method would perform more poorly for the later cases as the differences between the models increase.

Tables 2 and 3 present the results from the simulation study. They provide the number of times, out of 1,000 runs, in which the profit according to the NPI methods, as presented in this paper, are larger than for the corresponding classical method. Table 2 considers the probability of non-negative profit as optimality criterion, where $\underline{P}$, $\overline{P}$ and $H_{P,\omega}$ indicate that, for the NPI method, the lower probability, the upper probability, or the Hurwicz criterion was used, respectively. Table 3 considers the expected profit as optimality criterion, where $\underline{E}$, $\overline{E}$ and $H_{E,\omega}$ indicate that, for the NPI method, the lower expected profit, the upper expected profit, or the Hurwicz criterion was used, respectively.

As expected, the NPI methods perform worse than the classical methods in Case I, but for large n the performance of the NPI methods becomes good and the number of times it performs better than the classical method increases to close to 500. Note that the simulation results for Case II in Table 2 are identical to those for Case I. This follows from the fact that the classical optimal inventory levels that maximise the probability that the profit is greater than or equal to zero are equal for Gamma and Exponential distributions with the same mean. Furthermore, in Case II the NPI method performed relatively worse, when maximising the expected profit, than in Case I while the classical method does not have the benefit of using the exact same distribution as the one used for sampling the data. For Cases III–VI, the assumed model is not well aligned with reality, so the NPI method performs better than the classical method for all values of observations.

Table 2. Simulation results for lower and upper probabilities and their weighted function

	$n = 5$			$n = 50$			$n = 100$		
Case	$\underline{P}$	$\overline{P}$	$H_{P,\omega}$	$\underline{P}$	$\overline{P}$	$H_{P,\omega}$	$\underline{P}$	$\overline{P}$	$H_{P,\omega}$
I	384	385	384	436	436	436	486	486	486
II	384	385	384	436	436	436	486	486	486
III	551	557	551	704	704	704	710	710	710
IV	597	604	597	694	694	694	678	678	678
V	767	776	767	826	826	826	837	837	837
VI	822	831	822	889	889	889	896	896	896

Table 3. Simulation results for lower and upper expected profits and their weighted function

	$n = 5$			$n = 50$			$n = 100$		
Case	$\underline{E}$	$\overline{E}$	$H_{E,\omega}$	$\underline{E}$	$\overline{E}$	$H_{E,\omega}$	$\underline{E}$	$\overline{E}$	$H_{E,\omega}$
I	469	393	413	487	456	455	485	496	485
II	426	391	400	401	397	392	411	410	411
III	524	505	511	560	547	545	553	556	553
IV	622	615	610	676	679	676	714	715	714
V	751	685	705	733	725	722	726	726	726
VI	835	771	794	810	804	804	805	806	805

6. Concluding remarks

This paper has introduced the nonparametric predictive inference (NPI) approach to inventory problems. It enables inference based on only few model assumptions, hence it is strongly based on the data, which also implies that data, in the form of previous observed demands, must be available in order to be able to apply this method. The basic scenario of a single-period inventory problem with continuous random demand is considered. This paper presented how to find an optimal inventory level considering either the probability of getting a positive profit or the expected profit as optimisation criterion. Actually, the NPI approach quantifies uncertainty for inventory scenarios by the use of lower and upper probabilities for the profit corresponding to the demand for the next period, and corresponding upper and lower expected values of this demand. Further optimality criteria, taking weighted averages of corresponding NPI upper and lower probabilities or NPI upper and lower expected values of profit have also been considered.

The performances of the proposed NPI methods have been investigated via simulations. These show that the NPI methods tend to perform well for larger data sets, and outperform the classical method when assumptions underlying that method are not well aligned with the real world scenario.

From the perspective of computational complexity, the method presented in Section 4.1 is easy to implement and the complexity increases linearly with the number of demand observations. If this number is large, some reduction in computational effort can be achieved by implementing standard optimisation methods to search for the optimal solution. The methods in Section 4.2 are straightforward to implement with a single integer search, independent of the number of demand observations.

It is easily seen from the assumption $A_{(n)}$ for the future demand D_{n+1}, given in Equation 11, that for increasingly large numbers of demand data observations, the NPI methods' results converge to the corresponding results based on the empirical distribution, and therefore also to the results based on the unknown underlying distribution for the demands. It should further be noted that the imprecision, so the difference between upper and lower probabilities or expectations, decreases when the number of observations increases.

An important question is when the NPI method can or should be considered for inventory problems. Clearly, compared to the classical method with a fully known or assumed probability distribution for the random demand, the NPI method makes very few modelling assumptions and hence relies strongly on the available data. Hence, if such data are available, the NPI method is a good candidate to use for inventory decision support, unless one has good reasons to believe that a specific probability distribution describes the demand process adequately. As the simulations have shown, the NPI method cannot outperform the classical method if such beliefs are indeed correct, yet the results will get close for larger numbers of data. If, however, the assumed distribution is not in line with reality, the NPI method will have advantages. If one has little or no knowledge about the demand process, or explicitly does not want to use any such possible knowledge, then the NPI method is suitable for inventory decision support.

Whenever a new mathematical theory is presented for an important decision support scenario, there is a need to build on the first basic results in order to fit specific scenarios well. For inventory problems, a key aspect is inhomogeneity of demand over time, which goes beyond the basic setting with exchangeable demands considered in this paper. One obvious scenario is where the demand can be modelled as a time-series, with for example seasonality or other features. Recently, a start has been made in developing NPI generally for such scenarios, and its implementation to specific inventory problems is an interesting topic for future research. The value to managers of the current approach could also be questioned, in light of the early stages of the theory and hence the restriction to a basic setting. Nevertheless, in addition to the comments in the previous paragraphs, it would be recommendable to managers, who take inventory decisions seriously, to understand the available mathematical theories and methods to support their decisions, and to reach a balanced decision with the support of such methods where possible. Due to the limited mathematical assumptions underlying the NPI approach, the guidance provided by the method presented in this paper is likely to be of value, either directly or indirectly. Indirect use of the method is intended in the following sense: Suppose one has some observed demand data, and wishes to reach a decision for the inventory level for the next demand. If one considers the guidance provided by the NPI method, together with guidance provided by one or more other methods, and if the difference in recommended inventory levels is large, then any difference from the level recommended by the NPI method indicates the influence of stronger assumptions underlying the other methods used, as illustrated by the simulation study in Section 5. If a manager has good reason to support the stronger assumptions, than of course a method which uses these assumptions can be used, but even then considering the NPI-based method in parallel could be useful in order to understand to what extent the guidance is based on the data or on the model assumptions.

As this is the first study combining NPI with inventory problems, there are major opportunities to extend the theory and methodology, along the lines of many inventory scenarios considered in the literature. These include multi-period problems, different cost structures, constraints on inventory or order size, and so on. It is of course also of interest to consider inventories for multiple items. All such topics lead to exciting opportunities for future research, which may also require further theoretical development of the NPI approach.

Acknowledgements

The authors are grateful to the Associate Editor and reviewers whose comments led to improved presentation. Kholood Alyazidi acknowledges the financial support from King Saud University in Saudi Arabia and the Saudi Arabian Cultural Bureau (SACB) in London for facilitating her scholarship and PhD studies at Durham University.

Disclosure of interests

The authors have no interests to declare.

References

Arrow, K. J., & Hurwicz, L. (1977). An optimality criterion for decision making under ignorance. In: Arrow, K. J. and Hurwicz, L. (eds.). *Studies in Resource Allocation Processes,* Cambridge University Press, *pp*, 463–471.

Augustin, T., & Coolen, F. P. A. (2004). Nonparametric predictive inference and interval probability. *Journal of Statistical Planning and Inference*, *124*, 251–272.

Augustin, T., Coolen, F. P. A., de Cooman, G., & Troffaes, M. C. M. (2014). *Introduction to imprecise probabilities.* Wiley, Chichester.

Chen, J., Coolen, F. P. A., & Coolen-Maturi, T. (2019). On nonparametric predictive inference for asset and European option trading in the binomial tree model. *Journal of the Operational Research Society*, *70*, 1678–1691.

Chu, J., Huang, K., & Thiele, A. (2019). A robust optimization approach to model supply and demand uncertainties in inventory systems. *Journal of the Operational Research Society*, *70*, 1885–1899.

Collier, D. A., & Evans, J. R. (2010). *Operations management: Integrated goods and services approach.* Nelson Education, Canada.

Coolen, F. P. A. (1998). Low structure imprecise predictive inference for Bayes' problem. *Statistics & Probability Letters*, *36*, 349–357.

Coolen, F. P. A. (2006). On nonparametric predictive inference and objective Bayesianism. *Journal of Logic, Language and Information,* Springer, *15*, 21–47.

Coolen, F. P. A., & Coolen-Schrijner, P. (2003). A nonparametric predictive method for queues. *European Journal of Operational Research*, *145*, 425–442.

Coolen, F. P. A., & Yan, K. J. (2004). Nonparametric predictive inference with right-censored data. *Journal of Statistical Planning and Inference*, *126*, 25–54.

Coolen-Schrijner, P., Shaw, S. C., & Coolen, F. P. A. (2009). Opportunity-based age replacement with a one-cycle criterion. *Journal of the Operational Research Society*, *60*, 1428–1438.

De Finetti, B. (1974). *Theory of probability.* Wiley, London.

Geisser, S. (1993). *Predictive inference: An introduction.* Chapman and Hall, London.

Hadley, G. F., & Whitin, T. M. (1963). *Analysis of inventory systems.* Prentice-Hall, Englewood Cliffs, N.J., USA.

Harris, F. W. (1913). How many parts to make at once. *The Magazine of Management*, *10*, 135–136.

He, T., Coolen, F. P. A., & Coolen-Maturi, T. (2019). Nonparametric predictive inference for European option pricing based on the binomial tree model. *Journal of the Operational Research Society*, *70*, 1692–1708.

Hill, B. M. (1968). Posterior distribution of percentiles: Bayes' theorem for sampling from a population. *Journal of the American Statistical Association*, *63*, 677–691.

Hill, B. M. (1988). De Finetti's theorem, induction, and $A_{(n)}$ or Bayesian nonparametric predictive inference (with discussion). In J.M. Bernardo, et al. (eds.). *Bayesian Statistics,* Oxford University Press, *3*, 211–241.

Kao, C., & Hsu, W.-K. (2002). A single-period inventory model with fuzzy demand. *Computers & Mathematics with Applications*, *43*, 841–848.

Karteek, P. R., & Jyoti, K. (2014). Deterministic and probabilistic models in inventory control. *International Journal of Engineering Development and Research,* IJEDR, *2*, 3100–3105.

Lawless, J. F., & Fredette, M. (2005). Frequentist prediction intervals and predictive distributions. *Biometrika*, *92*, 529–542.

Reid, R. D., & Sanders, N. R. (2007). *Operations management: An integrated approach.* Wiley, Hoboken.

Sankarasubramanian, E., & Kumaraswamy, S. (1983). Note on "optimal ordering quantity to realize a pre-determined level of profit". *Management Science*, *29*, 512–514.

Shafer, G. (1976). *A mathematical theory of evidence.* Princeton University Press, New Jersey.

Shih, W. (1979). A general decision model for cost-volume-profit analysis under uncertainty. *The Accounting Review*, *54*, 687–706.

Taha, H. A. (2017). *Operations research: An Introduction.* Pearson Education, England.

Van Hees, R. N., & Monhemius, W. (1972). *Production and inventory control: Theory and practice.* Macmillan International Higher Education, London.

Walley, P. (1991). *Statistical reasoning with imprecise probabilities.* Chapman and Hall, London.

Waters, D. (2008). *Inventory control and management.* Wiley, Chichester.

Yao, J. S., Chang, S. C., & Su, J. S. (2000). Fuzzy inventory without backorder for fuzzy order quantity and fuzzy total demand quantity. *Computers & Operations Research*, *27*, 935–962.